

TEAM KRYPTX · SIH PS-26164 · ENCRYPT. PROTECT.
TRUST.

CRYPTORA

Cryptographic Risk & Migration Intelligence — Project Report

What it is · Why it matters · How it works · What it is built with ·
How it differs · What was hard · How we answer the jury — with screenshots of every
step.

Version **1.1**

Date **21 Sep 2026**

Live demo **:5173 + :8000**

Deploy **Docker · one command**

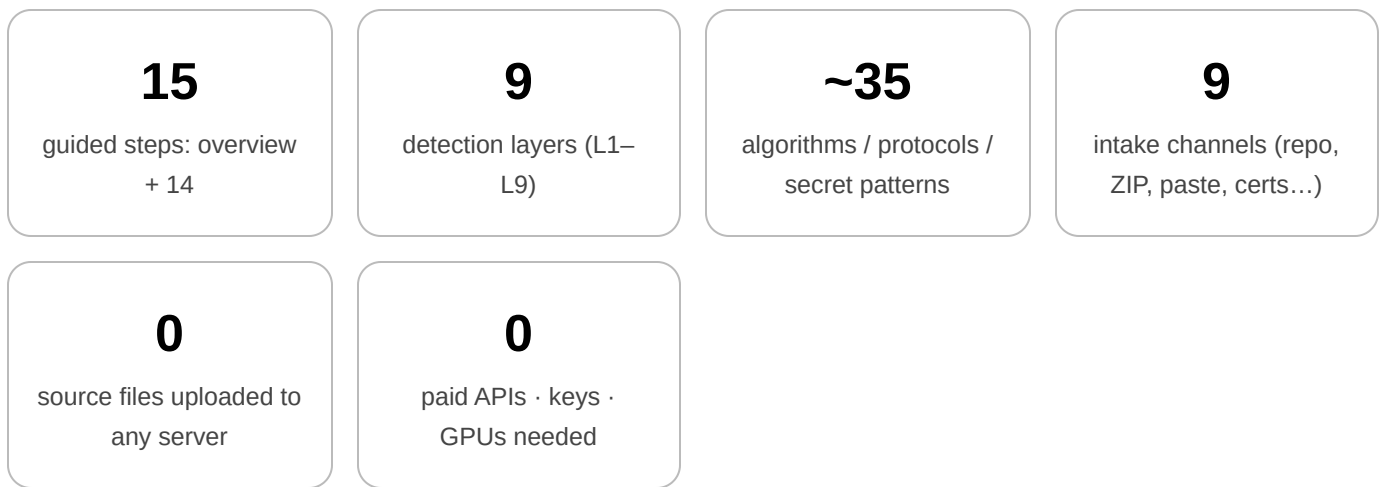
00 Contents

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01 What is CRYPTORA?

One line: CRYPTORA discovers where cryptography lives in your software, proves each finding with evidence, shows what quantum computers will break first, and plans a safe classical → hybrid → post-quantum migration — all running in your browser.

Thirty seconds: every organisation sits on years of RSA, ECC, AES, SHA, TLS and hardcoded secrets scattered across code, configs, containers and certificates — with no inventory. CRYPTORA scans real files, GitHub repos, pasted code, Dockerfiles, certificates and binaries; builds a Cryptographic Bill of Materials (CBOM); scores quantum + business risk with an interactive Mosca inequality; orders the break-first list; generates a NIST-aligned (ML-KEM / ML-DSA / SLH-DSA) migration roadmap; and exports board-ready reports. You cannot migrate what you cannot see — CRYPTORA makes crypto visible.



02 Why — the problem

1. Quantum computers break today's public-key crypto — it is a when, not an if

Shor's algorithm breaks RSA, ECC/ECDSA, DH and DSA once a sufficiently large cryptographically-relevant quantum computer exists; Grover's algorithm halves symmetric/hash strength (AES-128 → ~64-bit, SHA-256 → ~128-bit). NIST has already standardised the replacements — **ML-KEM (FIPS 203)**, **ML-DSA (FIPS 204)**, **SLH-DSA (FIPS 205)** — and

agencies (e.g. NSA CNSA 2.0) mandate PQC transition in national-security systems by the early 2030s. Industry timelines are converging on the same decade.

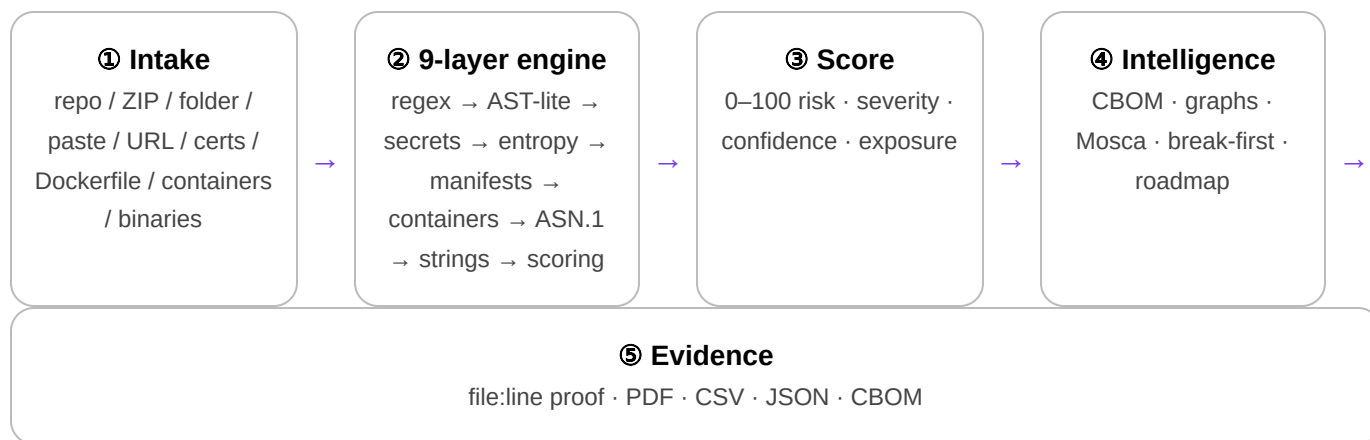
2. “Harvest now, decrypt later” makes waiting dangerous

Adversaries can record encrypted traffic and secrets *today* and decrypt them when quantum hardware arrives. Mosca’s inequality says it plainly: **if (data lifetime + migration time) > threat horizon, migration is already urgent**. Data with a 10-year shelf life is exposed *now* — CRYPTORA computes this interactively per organisation.

3. Nobody has a crypto inventory

Crypto sprawls: legacy RSA-1024 in one microservice, MD5 password hashes in another, an expired SHA-1 certificate, a hardcoded API key in a Dockerfile, DES in a vendor binary. Spreadsheets and tribal knowledge cannot track this. Without discovery + evidence + prioritisation, migration projects stall at step zero. That is the exact gap — discovery-to-roadmap for PQC — that CRYPTORA fills, aligned with SIH problem statement PS-26164.

03 How it works



The 9 detection layers

Layer	What it does	Example hit
-------	--------------	-------------

L1 · Regex	~35 cryptographic patterns: algorithms, modes, protocols, key sizes, JWT/TLS markers	<code>RSA.generate(1024)</code> , <code>AES.MODE_ECB</code> , <code>hashlib.md5</code>
L2 · AST-lite	Structure-aware call/import parsing for JS/TS & Python — boosts confidence beyond regex	import → call-graph link “this file really calls RSA”
L3 · Secrets	Hardcoded keys, API tokens, PEM blocks (values redacted on display)	<code>API_KEY = "sk-live-..."</code> (shown redacted)
L4 · Entropy	Shannon entropy per token + whole-file bytes → hidden/obfuscated crypto	High-entropy blob that regex cannot name
L5 · Manifests	npm / PyPI / Go / Cargo / Maven / Gradle parsing → crypto-relevant packages	<code>crypto-js@3.1.9</code> → capability map
L6 · Containers	Dockerfile static analysis: base images, installed crypto libs, leaked secrets	<code>libssl1.0</code> in image, <code>ENV SECRET=...</code>
L7 · Certificates	Hand-rolled ASN.1/DER parser: subject, issuer, validity, signature & key algorithm	SHA-1 cert expiring in 12 days
L8 · Binaries	Printable-string extraction + crypto marker search in binaries/documents — never executed	<code>DES_ecb_encrypt</code> inside a vendor .so
L9 · Risk + CBOM	Deterministic scoring, severity, exposure aggregation and CBOM assembly	risk 100 · CRITICAL · exposure 59/100

Scoring (deterministic — same scan, same numbers, every time)

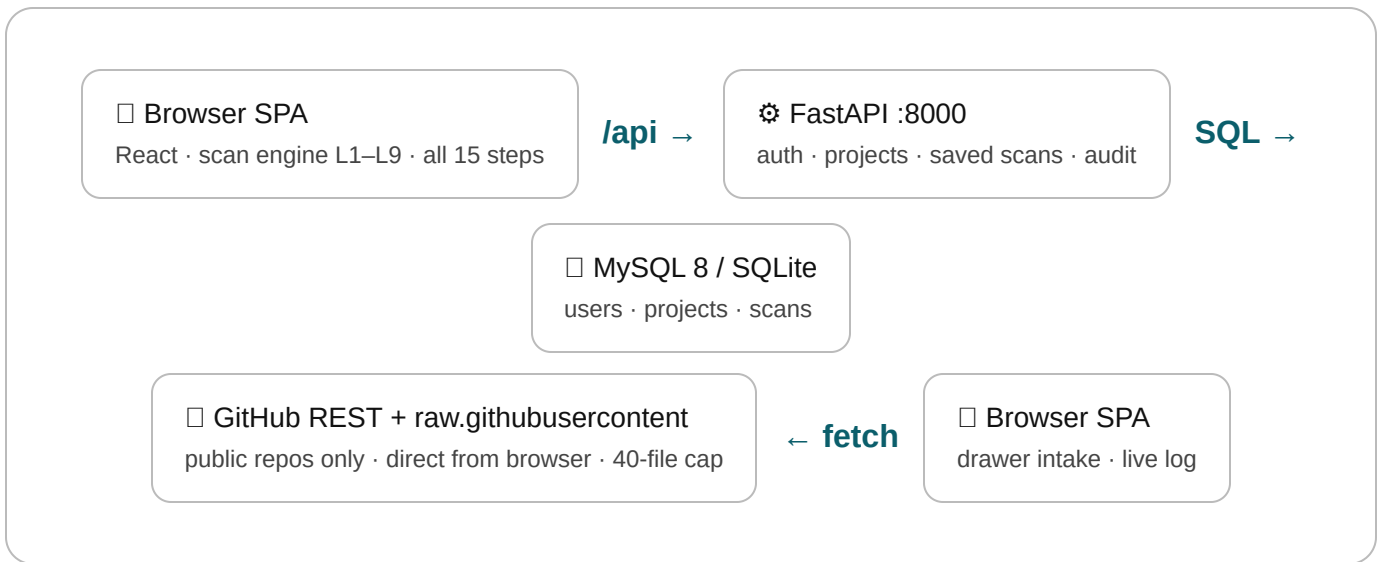
Each finding gets a **0–100 risk score** (algorithm weakness × key size × exposure × confidence), a **severity** (Critical/High/Medium/Low/Info), a **quantum flag** (Shor-breakable / Grover-weakened / safe / PQC) and a **confidence %**. Findings roll up into file, algorithm and portfolio exposure. Business context (data lifetime, migration time, threat horizon) feeds Mosca’s inequality and re-orders the break-first queue. The roadmap then maps each legacy primitive to its NIST PQC successor in phased waves.

Honesty rule enforced in code: with no scan, every page shows “No scan yet”. Empty result sets render “none found”. Secret *values* are redacted. The engine never invents rows to look impressive —

a claim the jury can verify live by scanning an empty file.

04 What is used (technology stack)

Layer	Technology (pinned)	Why this choice
UI + scan engine	React 18.3 + Vite 5 (plain JS, no TS build step)	Engine + UI in one codebase; instant dev loop; tiny static bundle (~400 KB JS) deployable anywhere
File intake	JSZip 3.10 (client-side)	ZIP/folder scans fully offline in-browser; nothing uploaded
Backend API	FastAPI + Uvicorn + Pydantic v2 (Python 3.12)	Typed, fast, auto OpenAPI docs at <code>/docs</code> ; auth + projects + scan metadata only
Auth	JWT (PyJWT-style) + bcrypt, Google OAuth verify, OTP hook	Email/password, Google, phone-OTP sign-in; dev-mode OTP echo for demos
Database	SQLAlchemy 2 + MySQL 8 (prod) / SQLite (dev)	One ORM, two backends: zero-config local file, durable prod DB with healthchecks + init SQL
Prod serving	nginx (SPA + <code>/api</code> proxy) + Docker Compose	Single-origin deployment kills CORS issues; one-command <code>deploy.sh</code>
Verification	Playwright-core + headless Chromium (E2E), curl module checks	Real-browser proof: scans, clicks, screenshots, zero-error runs



Cost of the entire stack: ₹0. No paid APIs, no model keys, no GPU, no per-seat SaaS. A 1-vCPU / 2-GB VM serves a classroom because scanning compute lives in each visitor's browser, not the server.

05 How CRYPTORA is different

Capability	CRYPTORA	Generic SAST (Semgrep/SonarQube-style)	Manual audit / spreadsheets	Black-box vendor scanners
Built for PQC migration	Yes — end to end	No — generic bugs	Ad-hoc	Rarely; quantum as upsell
Crypto inventory (CBOM)	Auto, CycloneDX- flavoured, exportable	Partial / plugin	Manual, rots fast	Often proprietary format
Evidence per finding	file:line + matched code + detector	Usually yes	Yes but slow	Sometimes hidden
Quantum break-first	Shor/Grover- aware queue	No	Expert-only	Sometimes

ordering				
Mosca urgency math	Interactive, adjustable	No	Whiteboard	No
Classical → hybrid → PQC roadmap	Generated from the scan	No	Consulting engagement	Report add-on
Privacy: source never uploaded	Yes — in-browser engine	Varies (cloud tiers upload)	Yes	Usually uploads code
Deterministic, reproducible	Same scan → same numbers	Mostly	No	Opaque
Cost to adopt	Free / open-stack demo	Free—₹₹	₹₹₹ (expert time)	₹₹₹ (licenses)

The one-sentence differentiator for the jury: generic scanners find *bugs* — CRYPTORA builds the *crypto inventory*, proves it, and turns it into a *funded migration plan*, without your code ever leaving your machine.

06 Screenshots of every process

Captured from the live build (`:5173` + `:8000`) by an automated real-browser run: landing → upload → live scan → all 14 intelligence steps → sign-in. Zero JavaScript errors across the run. Each image below is the actual product, not a mockup.



1

CRYPTORA ENCRYPT · PROTECT · TRUST How it works Platform PQC Trust Contact Sign in Launch App →

CRYPTOGRAPHIC RISK & MIGRATION INTELLIGENCE · TEAM KRYPTX · SIH PS-26164

Find every secret your code is keeping from you.

CRYPTORA discovers where cryptography lives in your software, proves each finding with evidence, shows what quantum computers break first, and plans a safe classical → hybrid → PQC migration.

🔍 Scan your code now Sign in / Create account

✓ runs in your browser ✓ no source ever uploaded* ✓ CycloneDX CBOM ✓ FIPS 203/204/285 paths

1. Discover → 2. Evidence → 3. CBOM → 4. Risk → 5. Prioritize → 6. Recommend → 7. Migrate → 8. Validate

🔍 Crypto Discovery

~30 algorithms, modes & protocols. Regex + AST-lite parsing, hardcoded-secret & entropy sweeps. Every hit carries file, line, detector & confidence.

📋 Evidence + CBOM

CycloneDX-flavoured software bill of materials generated from real scan evidence – view, filter, export. Nothing invented.

🔗 Dependency Graph

Manifest parsing across npm, PyPI, Go, Cargo, Maven & more. Crypto-relevant packages traced to capability.

🔮 Quantum + Business Risk

⚡ What Will Break?

→ Hybrid Migration

1. Landing page — public entry: pitch, trust badges (in-browser, no upload, CycloneDX, FIPS paths), and the 8-stage pipeline strip.

CRYPTORA
CRYPTO-INTELLIGENCE

Executive Overview
Live results from your scan

CRYPTORA // EXECUTIVE OVERVIEW

Cryptographic Risk & Migration Intelligence

CRYPTORA discovers where cryptography exists in your code, proves each finding with evidence, prioritizes what quantum computers will break first, and simulates a safe classical/hybrid/PQC migration path – all on today's classical computers.

Discover → Evidence + CBOM → Dependency Graph → Quantum + Business Risk → Impact

No scan yet
Upload real files, a public GitHub repo, or paste code to analyze.
Every stat, chart, and finding on this page is computed. Nothing here is sample data.

Upload / Analyze

CRYPTORA does not invent new cryptography – it applies NIST-standardized PQC (ML-KEM / FIPS 203, ML-DSA / FIPS 204) hardware is required to run this platform. This build is a heuristic, regex + AST-lite static scanner running AST/binary/dependency analysis pipeline. Team CRYPTA - 336 PG-20364 - Encrypt. Protect. Trust.

Upload & Analyze

Scan real files, a public GitHub repo, a pasted snippet, certificates, Dockerfiles, containers, or binaries/documents – live, in your browser. **Nothing leaves your machine** except direct GitHub/raw fetches you request.

Files / ZIP Folder **Repository URL** Paste Code

Link / URL Certificates Dockerfile Containers

Binaries / Docs

☒ Skip lockfiles (recommended – kills integrity-hash noise)
☒ Skip minified / build output

PUBLIC GITHUB REPOSITORY URL
https://github.com/owner/repo

BRANCH (OPTIONAL – DEFAULTS TO THE REPO'S DEFAULT BRANCH)
main

Fetches directly from GitHub's public REST API and raw.githubusercontent.com – nothing passes through a CRYPTORA server. Public repos only, up to 40 files, subject to GitHub's unauthenticated rate limit (60 requests/hour/IP).

Start Analysis

What this scan actually does

- L1+L2:** regex for ~35 crypto algorithms & protocols, plus real **AST-lite parsing** for JavaScript & Python
- L3:** hardcoded key / secret / PEM-block detection (values redacted)
- L4:** Shannon entropy scan – token-level + whole-file bytes
- L5-L8:** manifest parsing, Dockerfile analysis, real **ASN.1 certificate parsing**, binary/document static inspection – never executes anything

ESTATE MONITORING | LIVE

2. Upload & Analyze drawer — 9 intake tabs (Files/ZIP, Folder, GitHub repo, Paste Code, Link, Certificates, Dockerfile, Containers, Binaries/Docs) with noise-exclusion toggles and an honest “what this scan actually does” panel.

The screenshot displays the Cryptora Crypto Discovery Engine interface. On the left is a sidebar with a navigation menu including: Executive Overview, **Crypto Discovery** (highlighted), CBOM & Evidence, Dependency Graph, Certs, Containers, Bin., Quantum & Business Risk, Crypto Agility, What Will Break?, Attack Surface, Migration Roadmap, Validation & Rollback, Crypto Drift, AI Crypto Investigator, Hidden Encryption, and Reports. The main area is titled "Crypto Discovery" with the subtitle "Scan real files, a GitHub repo, or pasted code". It features buttons for "Upload / Analyze", "Pasted code", and "Sign in", along with a timestamp "23:04:43". Below the title, it says "STEP 1 // DISCOVERY" and "Crypto Discovery Engine". A description states: "Scans whatever you point it at – for ~35 cryptographic algorithms, protocols, and hardcoded-secret patterns, entirely in your browser." The central "Live Scan" section shows a terminal output for a scan of "Pasted code". The output details the engine layers (L1 to L8), skipping of dependency lockfiles and minified/build artifacts, scanning progress (1/1), and L9 scoring. Below the terminal, there are hit chips for each layer: L1: 3, L2: 3, L3: 2, L4: 0, L5: 0, L6: 0, L7: 0, L8: 0. On the right, a "Detection Methods" sidebar lists various techniques: REGEX PATTERNS (~35 rules covering RSA, ECC, etc.), AST-LITE PARSING (structured call + import for JavaScript/TypeScript), SECRET DETECTION (flags embedded PEM private blocks), ENTROPY ANALYSIS (Shannon entropy per token), and MANIFESTS + CONTAINERS (parsing Dockerfile, ASN.1 certificates, etc.). At the bottom, a "Findings (8/8)" section is partially visible.

3. Step 1 · Crypto Discovery — the live scan dashboard. The terminal streams every engine phase (fetch → L1–L8 detectors → L9 scoring → CBOM) with per-layer hit chips below. The app jumps here automatically the moment a scan starts.

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CRYPTO INTELLIGENCE

Executive Overview

Crypto Discovery

CBOM & Evidence

Dependency Graph

Certs, Containers, Bin.

Quantum & Business Risk

Crypto Agility

What Will Break?

Attack Surface

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Crypto Drift

AI Crypto Investigator

Hidden Encryption

Reports

Crypto Discovery

Scan real files, a Github repo, or pasted code

Upload / Analyze

Pasted code

Sign in

23:04:44

Search algorithm, file, symbol

All severities

All categories

Findings JSON

ID	SEVERITY	FINDING	FILE	DETECTOR	RISK
F-0005	CRITICAL	AES symmetric	vuln-demo.py:7 AES.new	Regex + AST L2 97%	100 Critical
F-0008	CRITICAL	MD5 hash	vuln-demo.py:5 hexdigest	Regex + AST L2 97%	100 Critical
F-0001	HIGH	RSA + quantum-vulnerable asymmetric	vuln-demo.py:2 (match)	Regex L1 82%	87 Critical
F-0002	HIGH	RSA + quantum-vulnerable asymmetric	vuln-demo.py:6 RSA.generate	Regex + AST L2 97%	87 Critical
F-0011	HIGH	Hardcoded secret assignment secret	vuln-demo.py:4 (secret material)	Secret-scan L3 85%	85 Critical
F-0012	MEDIUM	Suspicious key-like literal secret	vuln-demo.py:4 (secret material)	Secret-scan L3 85%	55 Medium
F-0004	MEDIUM	AES symmetric	vuln-demo.py:3 (match)	Regex L1 82%	45 Medium
F-0010	INFO	Post-quantum (ML-KEM / Kyber) pqc	vuln-demo.py:8 (match)	Regex L1 82%	5 Info

CRYPTORA does not invent new cryptography – it applies NIST-standardized PQC (ML-KEM / FIPS 203, ML-DSA / FIPS 204, SLH-DSA / FIPS 205) around what it finds. No quantum hardware is required to run this platform. This build is a heuristic, regex + AST-lite static scanner running entirely in your browser – not a substitute for a full AST/binary/dependency analysis pipeline. Team KRYPTX · SIM PS-26164 · Encrypt. Protect. Trust.

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4. Step 1 findings table — every hit carries an ID, severity, algorithm, file:line, detector layer and risk score; clicking an ID opens the evidence drawer with the exact matched code.



5. Executive Overview — glass stat cards, the risk curve (every finding ranked by score), findings-by-file hotspot bars, severity donut and per-layer engine bars. Every number is computed from the live scan — nothing is sample data.

CRYPTORA
CRYPTO INTELLIGENCE

Executive Overview

1. Crypto Discovery

2. **CBOM & Evidence**

3. Dependency Graph

4. Certs, Containers, Bin.

5. Quantum & Business Risk

6. Crypto Agility

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9. Migration Roadmap

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12. AI Crypto Investigator

13. Hidden Encryption

14. Reports

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CBOM & Evidence
Every real finding, with file/line evidence

Upload / Analyze Pasted code Sign in 23:04:47

STEP 2 // CBOM & EVIDENCE

Cryptographic Bill of Materials

Raw JSON CBOM (CycloneDX) SBOM (CycloneDX)

Every real match from your scan, with exact file, line, and matched text. No black box.

Search findings...

All Severity All Categories

File	Category	Severity	Quantum Risk	Detector / Layer	Confidence
vuln-demo.py:7	AES	CRITICAL	NONE		
vuln-demo.py:5	MD5	CRITICAL	NONE		
vuln-demo.py:2	RSA	HIGH	HIGH		
vuln-demo.py:6	RSA	HIGH	HIGH		
vuln-demo.py:4	Hardcoded secret assignment	HIGH	NONE		
vuln-demo.py:4	Suspicious key-like literal	MEDIUM	NONE		
vuln-demo.py:3	AES	MEDIUM	NONE		
vuln-demo.py:8		SAFE			

AES
symmetric · Quantum: NONE

MD5
hash · Quantum: NONE

RSA
asymmetric · Quantum: HIGH

RSA
asymmetric · Quantum: HIGH

Hardcoded secret assignment
secret · Quantum: NONE

Suspicious key-like literal
secret · Quantum: NONE

AES
symmetric · Quantum: NONE

SAFE

AES
Category: symmetric
Line: 7
Quantum risk: NONE
Severity: CRITICAL
Detector / Layer: Regex + AST · L2 · AST-Lite
Confidence: 97%

MATCHED TEXT

```
cipher = AES.new(b"0123456789abcdef", AES.MODE_ECB)
```

LINE 7 IN CONTEXT

```
5: password_hash = hashlib.md5(b"password123").hexdigest()
6: key = RSA.generate(1024)
7: cipher = AES.new(b"0123456789abcdef", AES.MODE_ECB)
8: import ml_kem # ML-KEM-768 hybrid plan
9:
```

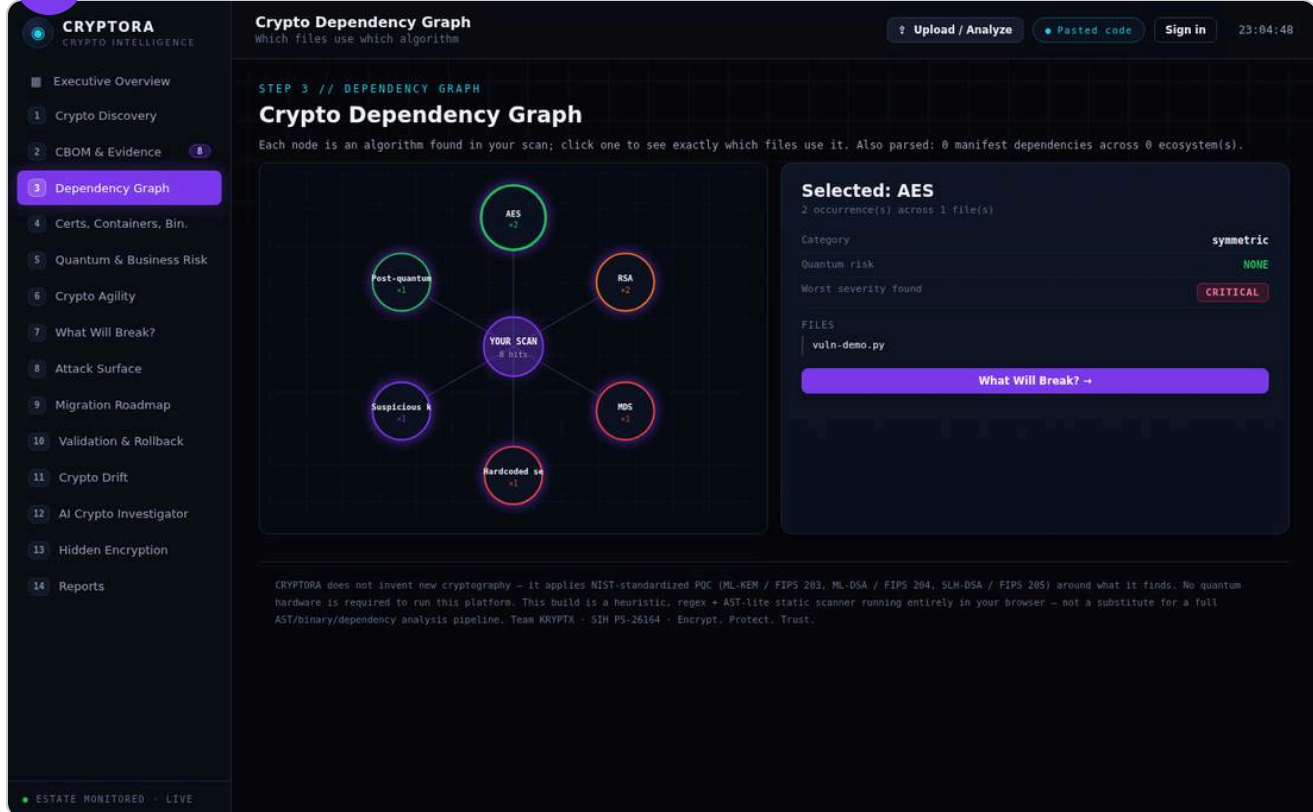
Explainable Risk Score
recomputed live from this finding + context — not a fixed number

DATA SENSITIVITY: Internal
BUSINESS CRITICALITY: Medium

7.8 / 10 P1

FACTOR BREAKDOWN
Algorithm/severity baseline

6. Step 2 · CBOM & Evidence — a CycloneDX-flavoured Cryptographic Bill of Materials generated from real findings, with per-finding matched text, line context and an Explainable Risk Score whose diverging factor bars recompute live — plus JSON export for auditors and toolchains.



7. Step 3 · Dependency Graph — glowing radial map of algorithms found in the scan on a grid backdrop. Clicking a node shows occurrences, category, quantum risk, worst severity and the exact files using it.

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CRYPTO INTELLIGENCE

Executive Overview

1. Crypto Discovery
2. CBOM & Evidence
3. Dependency Graph
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5. Quantum & Business Risk
6. Crypto Agility
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ESTATE MONITORED · LIVE

Certs, Containers, Bin.
X.509 · Dockerfile · Binaries

Upload / Analyze Pasted code Sign in 23:04:49

STEP 4 // CERTS, CONTAINERS, BINARIES

Certs, Containers, Bin.

Real ASN.1 parsing (signature OID, validity, key-size heuristic) + Dockerfile static analysis. Private key bytes are never displayed.

CERTIFICATES	EXPIRED / WEAK	CONTAINER SIGNALS	BINARIES INSPECTED
0	0	0	0

Certificates 0 parsed

No certificate blocks found in this scan.

Container signals 0 notes

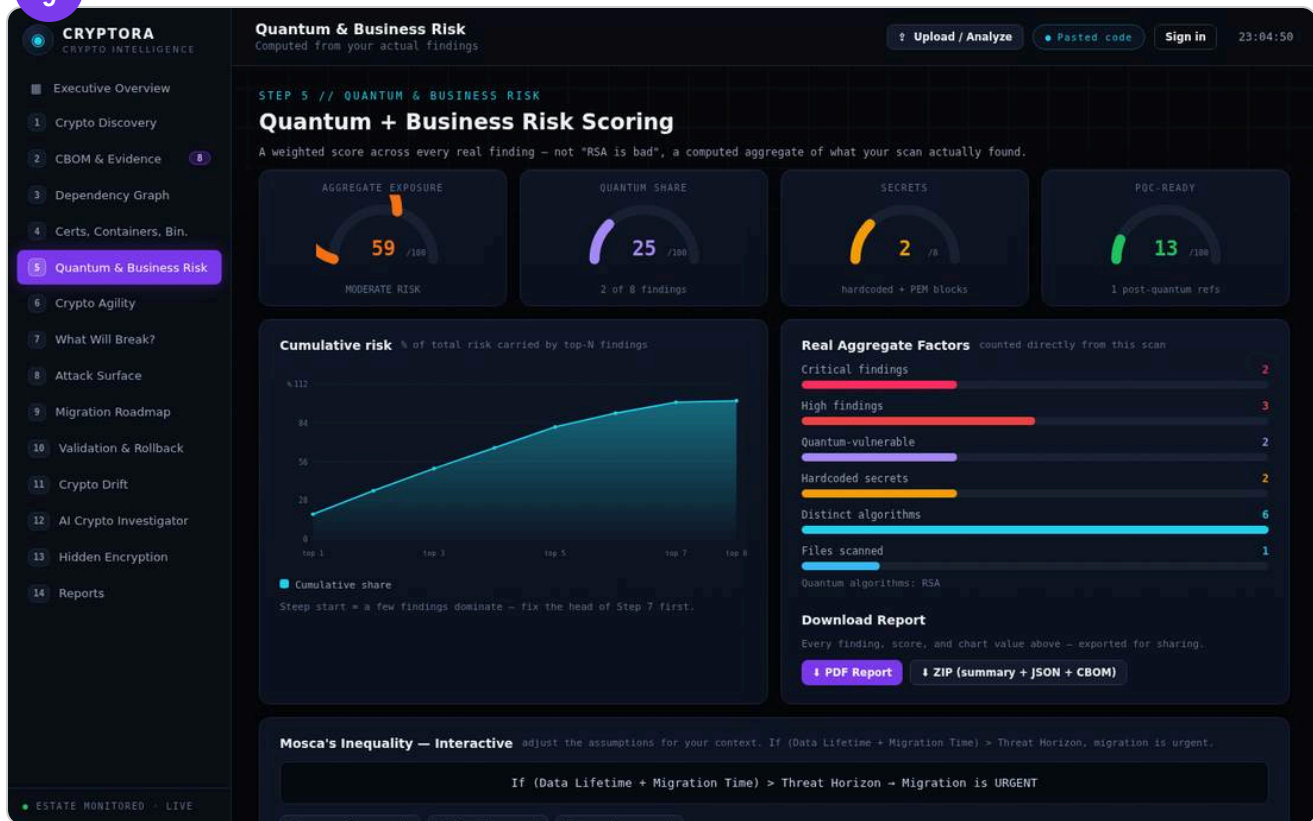
No Dockerfile/compose signals in this scan.

Binaries / documents 0 statically inspected

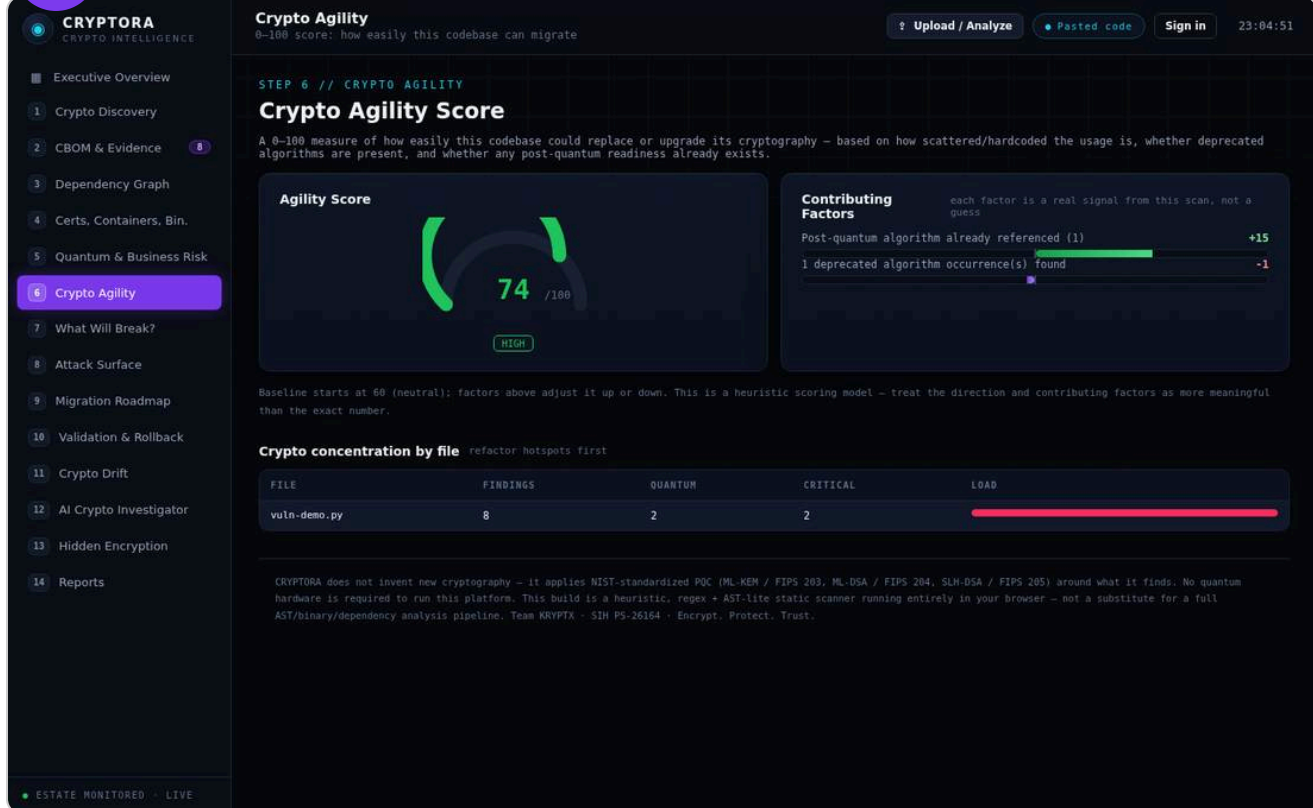
No binaries/documents in this scan.

CRYPTORA does not invent new cryptography – it applies NIST-standardized PQE (ML-KEM / FIPS 203, ML-DSA / FIPS 204, SLH-DSA / FIPS 205) around what it finds. No quantum hardware is required to run this platform. This build is a heuristic, regex + AST-lite static scanner running entirely in your browser – not a substitute for a full AST/binary/dependency analysis pipeline. Team KRYPTX · SIH PS-26164 · Encrypt. Protect. Trust.

8. Step 4 · Certs, Containers & Binaries — real X.509/ASN.1 certificate parsing (subject, issuer, validity, key algorithm), Dockerfile crypto analysis and static binary/document string inspection. Empty scans render honest “none found” states, never invented rows.



9. Step 5 · Quantum & Business Risk — four-dial gauge row (aggregate exposure, quantum share, secrets, PQC-ready), cumulative-risk curve, Real Aggregate Factors bars, PDF/ZIP export, and an interactive Mosca calculator with sliders (data lifetime + migration time vs threat horizon) and presets.



10. Step 6 · Crypto Agility — a 0–100 score for how easily this codebase can migrate, with diverging +/- factor bars showing exactly what raises or lowers it, plus a per-file crypto-concentration table of refactor hotspots.

CRYPTORA
CRYPTO INTELLIGENCE

Executive Overview

- Crypto Discovery
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- Quantum & Business Risk
- Crypto Agility
- What Will Break?**
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- Reports

ESTATE MONITORED · LIVE

What Will Break?

Break-first ordering

Upload / Analyze Pasted code Sign in 23:04:52

STEP 7 // WHAT WILL BREAK?

Break-first order = risk score + Mosca gap pressure (Z=12y, gap +1y) on quantum-vulnerable items.

Rank	Item	Severity	Chips	Risk Score	Urgency
1	AES F-0005 · vuln-demo.py:7	CRITICAL	critical severity	100	108
2	MD5 F-0008 · vuln-demo.py:5	CRITICAL	critical severity	100	108
3	RSA F-0001 · vuln-demo.py:2	HIGH	quantum + Mosca-urgent	87	91
4	RSA F-0002 · vuln-demo.py:6	HIGH	quantum + Mosca-urgent	87	91
5	Hardcoded secret assignment F-0011 · vuln-demo.py:4	HIGH	exposed secret	85	85
6	Suspicious key-like literal F-0012 · vuln-demo.py:4	MEDIUM	exposed secret	55	55
7	AES F-0004 · vuln-demo.py:3	MEDIUM		45	45
8	Post-quantum (ML-KEM / Kyber) F-0018 · vuln-demo.py:8	INFO		5	5

Why First #1 - F-0005

AES

vuln-demo.py:7 · risk 100 · urgency 108

CRITICAL

critical severity

risk score 100

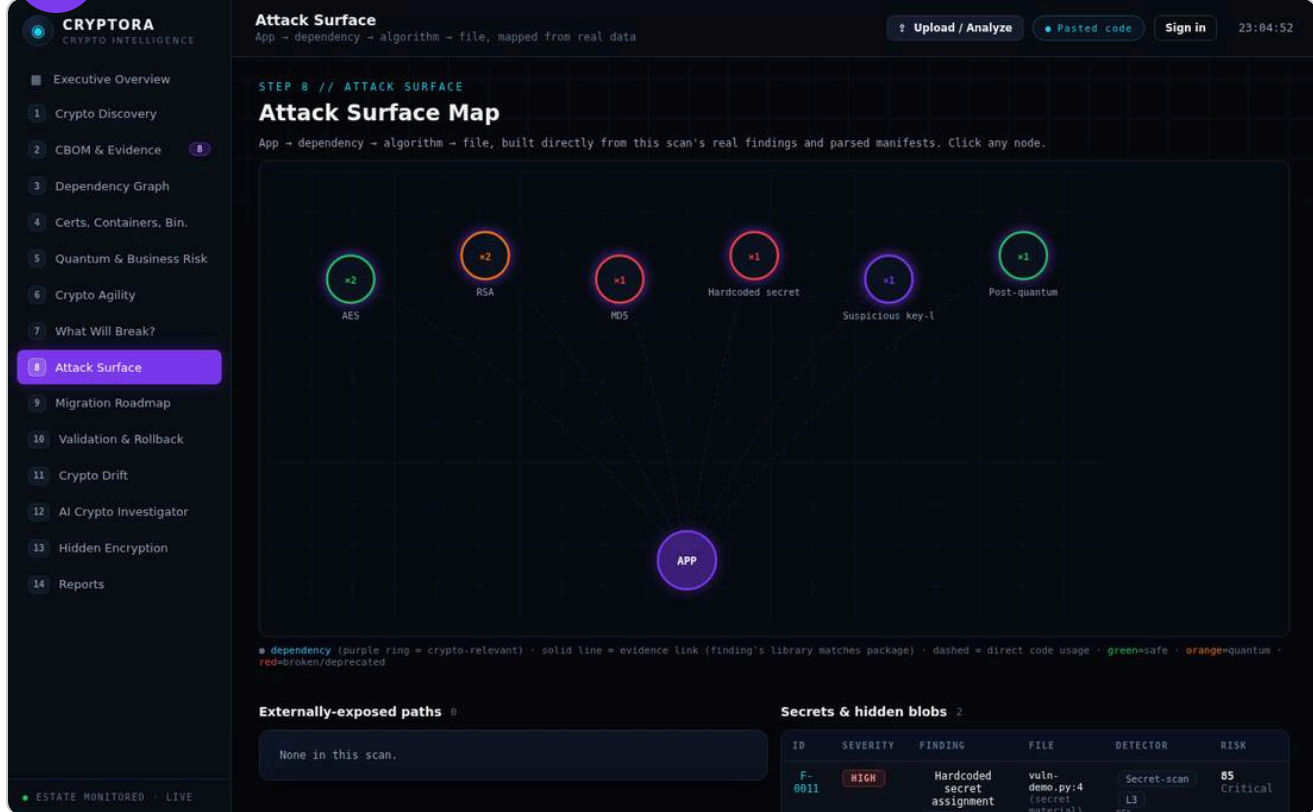
EVIDENCE

```
cipher = AES.new(b"0123456789abcdef", AES.MODE_ECB)
```

AES → — (symmetric; no hybrid needed) → AES-256-GCM (Grover-resistant at 256-bit)

Open full evidence

11. Step 7 · What Will Break? — break-first ranked list with medals, urgency scores and reason chips (quantum + Mosca-urgent, exposed secret, critical severity), alongside a Why First panel showing evidence and the migration path for the selected item.



12. Step 8 · Attack Surface Map — glowing app → dependency → algorithm → file graph with curved evidence links, built from real findings and parsed manifests, with externally-exposed paths and a secrets/hidden-blob table underneath.

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AI Crypto Investigator

Hidden Encryption

Reports

Migration Roadmap

Classical → Hybrid → PQC

Upload / Analyze

Pasted code

Sign in

23:04:53

STEP 9 // MIGRATION ROADMAP

Migration Roadmap

Phases auto-generated from your risk ranking. Status persists locally per finding. Classical → Hybrid → PQC, aligned to FIPS 203/204/205.

All

ML-KEM

ML-DSA

SLH-DSA

Other

filter phases by PQC target family

Phase 1 — Now (0-30 days) (4 items)

F-0005 · AES vuln-demo.py:7 AES → (symmetric: no hybrid needed) → AES-256-GCM (Grover-resistant at 256-bit)

Discovered

F-0008 · MD5 vuln-demo.py:5 MD5 → SHA-384/SHA-512 or SHA3-256 / BLAKE2b

Discovered

F-0001 · RSA vuln-demo.py:2 RSA → RSA-2048+ + ML-KEM-768 (hybrid key exchange) → ML-KEM-768 / ML-KEM-1024 (FIPS 203)

Discovered

F-0002 · RSA vuln-demo.py:6 RSA → RSA-2048+ + ML-KEM-768 (hybrid key exchange) → ML-KEM-768 / ML-KEM-1024 (FIPS 203)

Discovered

Phase 2 — Next (1-3 months) (1 items)

F-0011 · Hardcoded secret assignment vuln-demo.py:4 Embedded secret/key → Vault/KMS + rotation (never embed)

Discovered

Phase 3 — Later (3-12 months) (3 items)

F-0012 · Suspicious key-like literal vuln-demo.py:4 Embedded secret/key → Vault/KMS + rotation (never embed)

Discovered

F-0004 · AES vuln-demo.py:3 AES → (symmetric: no hybrid needed) → AES-256-GCM (Grover-resistant at 256-bit)

Discovered

F-0010 · Post-quantum (ML-KEM / Kyber) vuln-demo.py:8 Post-quantum (ML-KEM / Kyber) → (already PQC) → Keep ML-KEM-768/1024 (FIPS 203)

Discovered

Status board all tracked findings

DISCOVERED

8

100%

PRIORITIZED

0

0%

PLANNED

0

0%

IN PROGRESS

0

0%

MIGRATED

0

0%

VALIDATED

0

0%

ESTATE MONITORED · LIVE

13. Step 9 · Migration Roadmap — ML-KEM | ML-DSA | SLH-DSA target-family filter, phased classical → hybrid → PQC plan generated from the scan, and a Status board with per-stage counts, percentages and a persistent kanban.

CRYPTORA
CRYPTO INTELLIGENCE

Executive Overview

Crypto Discovery

CBOM & Evidence

Dependency Graph

Certs, Containers, Bin.

Quantum & Business Risk

Crypto Agility

What Will Break?

Attack Surface

Migration Roadmap

10 Validation & Rollback

Crypto Drift

AI Crypto Investigator

Hidden Encryption

Reports

ESTATE MONITORED · LIVE

Validation & Rollback
Rescan proof + rollback

Upload / Analyze Pasted code Sign in 23:04:54

STEP 10 // VALIDATION & ROLLBACK

Validation & Rollback

Validate migrations by rescanning and diffing. Rollback plans are generated per finding from its migration path.

Current scan: scan-muburkao · 8 findings
Baseline snapshot: – none saved yet (see Crypto Drift) –

Rollback plan generator top 8 risks

F-0005 · AES · vuln-demo.py:7

1. Snapshot: pin current AES behaviour + keys (backup, don't delete).
2. Ship hybrid behind a feature flag (– (symmetric: no hybrid needed)) – verify both paths in staging.
3. Roll forward to AES-256-GCM (Grover-resistant at 256-bit); keep flag for instant rollback within one deploy.
4. Rescan with CRYPTORA – this page proves the finding is resolved.

F-0006 · MD5 · vuln-demo.py:5

1. Snapshot: pin current MD5 behaviour + keys (backup, don't delete).
2. Ship hybrid behind a feature flag – verify both paths in staging.
3. Roll forward to SHA-384/SHA-512 or SHA3-256 / BLAKE2b; keep flag for instant rollback within one deploy.
4. Rescan with CRYPTORA – this page proves the finding is resolved.

F-0001 · RSA · vuln-demo.py:2

1. Snapshot: pin current RSA behaviour + keys (backup, don't delete).
2. Ship hybrid behind a feature flag (RSA-2048+ + ML-KEM-768 (hybrid key exchange)) – verify both paths in staging.
3. Roll forward to ML-KEM-768 / ML-KEM-1024 (FIPS 203); keep flag for instant rollback within one deploy.
4. Rescan with CRYPTORA – this page proves the finding is resolved.

F-0002 · RSA · vuln-demo.py:6

1. Snapshot: pin current RSA behaviour + keys (backup, don't delete).
2. Ship hybrid behind a feature flag (RSA-2048+ + ML-KEM-768 (hybrid key exchange)) – verify both paths in staging.
3. Roll forward to ML-KEM-768 / ML-KEM-1024 (FIPS 203); keep flag for instant rollback within one deploy.
4. Rescan with CRYPTORA – this page proves the finding is resolved.

14. Step 10 · Validation & Rollback — rescan-to-prove-fix workflow: snapshots before/after remediation, diff of resolved vs introduced findings, and rollback guidance if a migration regresses.

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Snapshot comparison

Upload / Analyze

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STEP 11 // CRYPTO DRIFT

Crypto Drift

Snapshots persist in your browser (localStorage). Compare any two to see new / removed / changed crypto.

Snapshot name_

+ Save current scan as snapshot

Clear

SNAPSHOT	SAVED	FILES	FINDINGS	QUANTUM	CRITICAL
demo-baseline	9/21/2026, 11:04:56 PM	1	8	2	2

Snapshot trend

real history - one point per saved snapshot + current scan

Findings

Quantum

Critical

Baseline_

Target_

CRYPTORA does not invent new cryptography - it applies NIST-standardized PQC (ML-KEM / FIPS 203, ML-USA / FIPS 284, SLH-DSA / FIPS 205) around what it finds. No quantum hardware is required to run this platform. This build is a heuristic, regex + AST-lite static scanner running entirely in your browser - not a substitute for a full AST/binary/dependency analysis pipeline. Team KRYPTIX · SIH PS-26164 · Encrypt. Protect. Trust.

ESTATE MONITORED · LIVE

15. Step 11 · Crypto Drift — save snapshots, compare any two (new / removed / unchanged crypto), and watch the real snapshot-trend chart of findings, quantum and critical counts over time.

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AI Crypto Investigator

Grounded Q&A over your real findings

Upload / Analyze

Pasted code

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STEP 12 // AI INVESTIGATOR

AI Crypto Investigator

Ask about any cryptography term, or – once you've scanned something – about your actual findings. Answers are generated locally, grounded in your real scan when one exists; not a hosted LLM, and not canned.

What's the highest-risk finding?

How many critical findings are there?

What's the risk score and priority?

Which algorithms are quantum-vulnerable?

Any transitive npm dependencies?

What should I fix first?

Ask a question about your scan, or click a suggestion above.

Ask about any cryptography term, or – once you've scanned something – about your actual findings. Answers are generated locally, grounded in your real scan when one exists; not a hosted LLM, and not canned.

Ask about a term, or your scan once you've run one...

Ask

Methodology Notes

How does CRYPTORA classify quantum risk?

Public-key algorithms (RSA, ECC/ECDSA/ECDF, DSA, classical Diffie-Hellman) are broken by Shor's algorithm on a sufficiently large quantum computer, so they're marked HIGH. Symmetric algorithms (AES-256, ChaCha20) and hashes (SHA-2/3) only face Grover's quadratic speedup – safe at 256-bit strength. PQC algorithms (ML-KEM, ML-DSA, SLH-DSA) are marked SAFE.

How is the risk score computed?

Severity baseline + quantum (+1.5) + secret material (+1.5) + path exposure, tuned by your business context. The exact factors are shown under every score – nothing is a black box.

What are the limits?

Heuristic static analysis: regex + AST-lite, entropy, manifests, Dockerfile, ASN.1 certs, binary strings. It cannot see runtime behavior, obfuscated binaries' internals, or transitive dependencies beyond manifests. No 100% coverage is claimed.

CRYPTORA does not invent new cryptography – it applies NIST-standardized PQC (ML-KEM / FIPS 203, ML-DSA / FIPS 204, SLH-DSA / FIPS 205) around what it finds. No quantum

16. Step 12 · AI Crypto Investigator — grounded Q&A strictly over the scan's real findings (no hallucinations: every answer cites finding IDs), plus suggested investigation queries for analysts.

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Real Shannon-entropy scan of your files

Upload / Analyze

Pasted code

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STEP 13 // HIDDEN ENCRYPTION

Cryptography Nobody Registered

Standard pattern matching misses crypto that's deliberately or accidentally hidden. This is a genuine Shannon-entropy calculation over every scanned file – high entropy (>6.6 bits/byte) suggests encrypted, compressed, or key-like content, even with no algorithm keyword present.

HIGH-ENTROPY FILES

0

FILES ANALYZED

1

PEAK ENTROPY (BITS/BYTE)

5.49

DETECTION METHOD

REAL

computed, not lookup

No files crossed the high-entropy threshold in this scan.

Entropy Scan

every scanned file, ranked by Shannon entropy

5.49

..vuln-demo.py

This is a real, computed measurement – not a lookup table.

Token-level candidates

2 high-entropy strings + embedded secrets

ID	SEVERITY	FINDING	FILE	DETECTOR	RISK
F-0011	HIGH	Hardcoded secret assignment secret	vuln-demo.py:4 (secret material)	Secret-scan L3 85%	85 Critical
F-0012	MEDIUM	Suspicious key-like literal secret	vuln-demo.py:4 (secret material)	Secret-scan L3 85%	55 Medium

CRYPTORA does not invent new cryptography – it applies NIST-standardized PQC (ML-KEM / FIPS 203, ML-DSA / FIPS 204, SLH-DSA / FIPS 205) around what it finds. No quantum hardware is required to run this platform. This build is a heuristic, regex + AST-lite static scanner running entirely in your browser – not a substitute for a full AST/binary/dependency analysis pipeline. Toss KRYPTOS · 5TH PS-26164 · Encrypt · Protect · Trust.

17. Step 13 · Hidden Encryption — Shannon-entropy analysis surfacing obfuscated/embedded crypto: high-entropy tokens, suspicious blobs and whole-file byte analysis that regex alone would miss.

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14 Reports

ESTATE MONITORED · LIVE

Reports

Executive report · PDF · CSV

Upload / Analyze

Pasted code

Sign in

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STEP 14 // REPORTS

Reports

Executive report, printable PDF, and machine-readable exports – all generated from the live scan.

Print / Download PDF

Findings CSV

Report JSON

CBOM JSON

CRYPTORA

CRYPTOGRAPHIC RISK & MIGRATION INTELLIGENCE

EXECUTIVE REPORT

9/21/2026, 11:04:42 PM

scan-muburkao

Cryptographic Risk Report — cryptora-scan

Source: Pasted code · Files scanned: 1 · Duration: 0.0s · Team KRYPTX

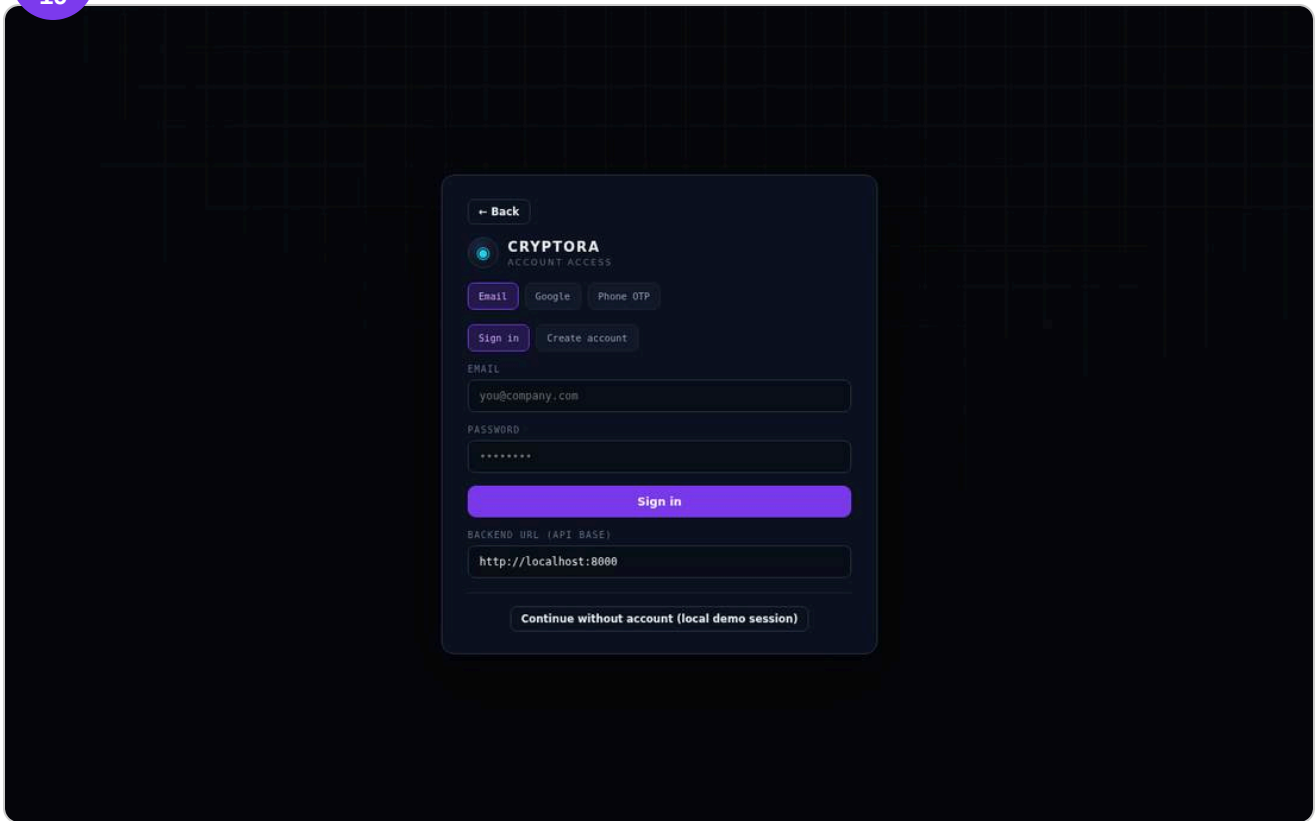
1. Executive summary

Total findings	8	Critical / High	2 / 3
Quantum-vulnerable	2	Embedded secrets	2
Hidden blobs	0	Certs / Deps parsed	0 / 0
Mosca (10+3 vs Z12)	ACT NOW – data outlives quantum-safe horizon (gap +1y)		

2. Top risks (by risk score)

#	ID	Severity	Finding	Location	Risk
1	F-0005	CRITICAL	AES	vuln-demo.py:7	100 (Critical)
2	F-0008	CRITICAL	MD5	vuln-demo.py:5	100 (Critical)
3	F-0001	HIGH	RSA [0]	vuln-demo.py:2	87 (Critical)
4	F-0002	HIGH	RSA [0]	vuln-demo.py:6	87 (Critical)
5	F-0011	HIGH	Hardcoded secret assignment	vuln-demo.py:4	85 (Critical)
6	F-0012	MEDIUM	Suspicious key-like literal	vuln-demo.py:4	55 (Medium)
7	F-0004	MEDIUM	AES	vuln-demo.py:3	45 (Medium)
8	F-0010	INFO	Post-quantum (ML-KEM / Kyber)	vuln-demo.py:8	5 (Info)

18. Step 14 · Reports — the board-ready executive report: summary table, top risks by score, quantum exposure & PQC path, with one-click Print/PDF, Findings CSV, Report JSON and CBOM JSON exports.



19. Sign in — email/password, Google OAuth and phone-OTP authentication with a visible, editable Backend URL (auto-detected per environment), unlocking cloud projects, saved scans and audit logs.

07 Difficulties faced & how we solved them

Q. False positives drowned real findings (lockfiles, minified bundles, build output).

A. Built an **exclusion engine**: lockfiles (`*-lock.json`, `*.lock`, `go.sum`) and minified/build artefacts are skipped with user-visible toggles and honest “excluded: N” counters in the live log. Precision first, then recall.

Q. Regex alone cannot tell `import RSA` (unused) from `RSA.generate()` (real usage).

A. Added **AST-lite**: lightweight structural call/import parsing for JS/TS and Python that confirms real usage and raises confidence (e.g. 97% on confirmed calls) — without shipping a full parser per language.

Q. Big repos could freeze a browser tab.

A. Bounded the work: 40-file GitHub cap with clear messaging, per-file timeouts, streaming progress into the live log, and ZIP/folder intake via JSZip so large private codebases scan offline without any upload.

Q. Secrets vs noise: how to flag keys without leaking them or crying wolf?

A. Layered signals (patterns + PEM structure + entropy + context) with severity tiers, and **values always redacted** in UI and exports. A finding says *where* and *what kind* — never the secret itself.

Q. Certificates and binaries with zero native dependencies?

A. Hand-rolled a minimal **ASN.1/DER reader** for X.509 fields and a printable-**strings extractor** with crypto-marker search for binaries/documents. Static inspection only — the engine **never executes** anything it scans.

Q. Backend URL broke across environments (localhost vs preview vs Docker vs public domain).

A. Implemented **environment auto-detection** (preview → derived API host, local → `localhost:8000`, prod → same-origin `/api`) plus a visible manual override on the Sign-in page. Verified in all four environments.

Q. Users felt the app was “stuck” after starting a scan (drawer blocked the view).

A. Reworked the flow: the drawer now **auto-closes and jumps to the live scan dashboard** the moment the engine starts, with a “Watch live” escape hatch mid-fetch. Verified by scripted browser tests.

Q. “Prove it’s really scanning” — the credibility problem every demo faces.

A. Three answers: (1) **deterministic engine** — same input, same finding IDs and scores, reproducible on stage; (2) **evidence drawers** — every row expands to matched code; (3) **hostile self-tests** — empty file → zero findings; `md5(x)` added → exactly one new F-ID. The jury can run these live.

08 Tackling jury questions

A · Product & purpose

Q. What problem are you solving, in one minute?

A. Organisations must migrate to post-quantum cryptography this decade but have no inventory of where crypto lives. CRYPTORA discovers it, proves it with evidence, scores quantum risk, and generates the migration plan. No inventory → no migration; we are step zero, done right.

► Demo move: [Overview numbers](#) → [CBOM evidence](#) → [Roadmap](#). 60 seconds.

Q. Who is the user? Who pays?

A. Users: AppSec engineers, auditors, CISOs. Payers: enterprises facing CNSA 2.0 / compliance deadlines and consultancies selling PQC-migration assessments. The demo is free; the roadmap monetises via team workspaces, CI gates and scheduled drift monitoring.

Q. Is this just a regex grep with a dashboard?

A. Regex is layer 1 of 9. AST-lite confirms real usage, entropy finds what regex cannot name, ASN.1 parses real certificates, manifests map dependencies — and L9 turns it into scored, ordered, exportable migration intelligence. A grep cannot produce a CBOM, a Mosca verdict or a phased PQC roadmap.

► Demo move: [Hidden Encryption page \(entropy blob regex missed\)](#) + [Reports PDF](#).

Q. What happens with a repo with no crypto at all?

A. Honest empty states: zero findings, “none found”, gauges at 0 — the engine never invents rows. This is a feature, and we demo it on stage with an empty file.

B · Technical depth

Q. How do you avoid false positives?

A. Four mechanisms: exclusion engine (lockfiles/minified/build), AST-lite usage confirmation, confidence scoring per finding, and severity tiers. Remaining noise is labelled low-confidence rather than hidden.

Q. How accurate is detection? Did you measure it?

A. Honest answer: this build is a heuristic static scanner; we validate deterministically (fixed fixtures → fixed finding IDs/scores) and ship demo-fixtures for repeatable checks. We do *not* claim ML-grade precision/recall numbers we haven't measured — the evidence drawer lets anyone audit every hit. Formal benchmarking against labelled corpora is explicit roadmap.

Q. Why in-browser instead of a server scanner?

A. Privacy (source never leaves the machine — huge for banks/healthcare), zero server cost per scan (scales to a classroom on a tiny VM), and offline capability (ZIP/folder scans with no network). The server stores only accounts and metadata.

Q. What about obfuscated, minified or compiled code?

A. Minified code is excluded from regex noise by default but still entropy-scanned; compiled artefacts go through the binary-strings layer; and Hidden Encryption surfaces high-entropy blobs regardless of readability. Full deobfuscation/AST-per-language is roadmap, stated openly.

Q. How does the 40-file GitHub cap affect real repos?

A. It bounds unauthenticated rate limits (60 req/hr/IP) and browser memory; the cap, skip counts and rate-limit errors are all shown honestly in the live log. Private/large codebases use ZIP or folder upload with no cap and no network. Authenticated GitHub + chunked fetching is roadmap.

C · Cryptography & PQC

Q. Which PQC algorithms do you recommend, and why?

A. Exactly the NIST standards — ML-KEM (FIPS 203) for key establishment, ML-DSA (FIPS 204) for signatures, SLH-DSA (FIPS 205) where stateless hash-based signatures fit — plus hybrid (classical + PQC) transition paths. We invent no cryptography; the footer of every page says so.

Q. Explain Mosca's inequality like I'm a CFO.

A. If your data must stay secret for D years, migration takes M years, and quantum arrives in T years — and $D+M > T$ — then attackers are already harvesting your data to decrypt later. Our calculator turns that into ACT NOW vs NOT YET URGENT with your own numbers.

► Demo move: Risk page → drag the three sliders → verdict flips live.

Q. Do you handle symmetric crypto and hashes, or only RSA/ECC?

A. All three families: asymmetric (RSA/ECC/DH/DSA + key sizes), symmetric (AES incl. ECB/CBC/GCM modes, DES/3DES, ChaCha20), hashes (MD5/SHA-1/SHA-2...), plus TLS/JWT/protocol markers and PQC references (already-migrated code is detected and celebrated, not flagged).

Q. What is a CBOM and why CycloneDX-flavoured?

A. A Cryptographic Bill of Materials — the machine-readable inventory auditors and tools consume. CycloneDX flavouring keeps exports compatible with the SBOM ecosystem (dependency-track-style pipelines) instead of a proprietary format.

D • Deployment, scale & security

Q. How do I run this? What does it cost?

A. Three ways: local (`npm run dev` + uvicorn, 5 min), Docker production (`./deploy/deploy.sh` , one command), or split static+API hosting. Cost: ₹0 in software; a 1-vCPU/2-GB VM suffices because scanning runs in visitors' browsers.

► Demo move: show the deployment guide + `/health` + `/docs` .

Q. Is it safe to scan proprietary code?

A. Yes by architecture: the engine runs locally, nothing is uploaded (except direct GitHub fetches you explicitly request), secrets are redacted, and scanned code is never executed. The backend only sees accounts and finding metadata.

Q. How do you store passwords and tokens?

A. bcrypt-hashed passwords, signed JWTs with configurable TTL, Google ID-token verification server-side, and secrets via environment (never committed). OTP is hook-ready with dev-mode echo for demos.

E · Vision & honesty

Q. What doesn't it do yet?

A. No full per-language AST, no dynamic analysis, no authenticated GitHub at scale, no team RBAC/CI gates yet — each is on the roadmap. We'd rather show a verified scope than overclaim; everything demoed today is real and reproducible.

Q. What would you build with 3 more months?

A. (1) Tree-sitter AST per language + SARIF export for CI; (2) GitHub App with chunked large-repo scans + PR comments; (3) team workspaces, scheduled drift monitors and Agility-trend dashboards; (4) labelled-corpus precision/recall benchmark published openly.

Q. Why should we pick you?

A. Working software over slides: 15 live steps, a real 9-layer engine, one-command deploy, and a demo the jury can attack — scan anything, watch the log, audit every finding, export the PDF. Plus honest engineering: we show our limits alongside our strengths.

09 Verification: proof the software works

Final acceptance run on the live build (headless Chromium driving the real UI, 21 Sep 2026):

Check	Method	Result
Full scan flow (paste → auto-jump → live log → findings → badge)	Scripted UI run	7/7 steps PASS
All 15 sidebar destinations navigate (empty state)	Real button clicks	15/15 PASS
All 15 sidebar destinations navigate (with scan data)	Real button clicks	15/15 PASS
JavaScript errors across all runs + 20 screenshots	Console + pageerror capture	0 errors
Production build	<code>npm run build</code>	✓ ~401 KB JS / ~23 KB CSS
Backend API	<code>/health</code> + <code>/api/config</code>	200 OK
Backend auto-detect (preview / local / docker / domain)	Mocked-location unit check	4/4 correct
Mobile (390px): hamburger → sidebar → scan → all pages	Real touch taps	PASS, 0 errors

Reproduce it: unzip the bundle → run backend + frontend (§3 of the Deployment Guide) → paste any crypto snippet → click through steps 1–14 with DevTools open. Deterministic engine: the same input always yields the same finding IDs and scores.

10 Deployment summary

Full instructions live in `CRYPTORA_Deployment_Guide.md` (same folder as this report). The three supported paths:

Path	Command	Serves
Local demo/dev	<code>uvicorn app.main:app --port 8000</code> + <code>npm run dev</code>	:5173 + :8000, SQLite, zero config
Docker production ★	<code>cp deploy/.env.example</code> <code>deploy/.env</code> → fill → <code>./deploy/deploy.sh</code>	:8080 (nginx+SPA+API proxy) + MySQL 8, healthchecks, volumes
VPS + HTTPS	Deploy path above + Caddy reverse proxy + domain A-record	Public <code>https://...</code> with auto-TLS

11 Roadmap & pitch script

Roadmap (next 3 horizons)

Now → 1 month	1 → 3 months	3 → 6 months
Tree-sitter AST per language · SARIF export · more fixtures · precision/recall benchmark	GitHub App (large repos, PR comments) · team workspaces + RBAC · scheduled drift monitors	CI policy gates · Agility trends · auditor multi-project rollups · hosted assessment tier

60-second pitch (say this)

“Quantum computers will break RSA and ECC – and attackers are harvesting encrypted data TODAY to decrypt later. But no company even knows where its cryptography lives. CRYPTORA finds it: [scan live] – every algorithm, every secret, every certificate, with file-and-line evidence. [CBOM] That’s your

crypto inventory. [Break-first] That's what quantum kills first. [Roadmap] And that's your NIST-approved migration plan – classical, to hybrid, to post-quantum. Nothing uploaded, nothing faked – scan anything yourself, right now. Team KRYPTX: Encrypt. Protect. Trust.”

5-minute demo order

Landing → Upload drawer → **live** paste scan → Discovery log → Overview numbers → CBOM evidence → Break-first → Roadmap → Reports PDF → Sign-in + saved scan. Keep a pre-scanned repo loaded as backup; if venue Wi-Fi dies, switch to ZIP upload and keep going.

CRYPTORA · Team KRYPTX · SIH PS-26164 · “Encrypt. Protect. Trust.”

Report v1.1 — 21 Sep 2026 (UI refresh: gauges, charts, ranked views, mobile) — screenshots: live build :5173+:8000 · See CRYPTORA_Deployment_Guide.md for operations.